CS 4412 Data Mining - M2 Initial Implementation

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03/05/2026

Introduction

This analysis explores patterns within the Student Performance dataset obtained from the UCI Machine Learning Repository. The dataset contains demographic, academic, social, and lifestyle attributes for secondary school students enrolled in Mathematics and Portuguese language courses.

The goal of this milestone is to perform exploratory data analysis (EDA), prepare the dataset through preprocessing and transformation, and apply an initial data mining technique to discover patterns in student behavior and academic performance.

The discovery questions guiding this analysis are:

1. Do family and socioeconomic attributes form recognizable patterns linked to academic consistency?
2. How do study habits and absences interact to shape academic performance patterns?
3. What natural clusters of students emerge when combining demographic, family, and academic features?

To begin answering these questions, the analysis will first explore the structure and quality of the dataset then prepares a set of meaningful variables before applying clustering techniques to identify natural student groups.

Data Loading

```
# Load both datasets from GitHub repository
student_math <- read.csv(
↪  "https://raw.githubusercontent.com/Colochoo/cs4412-project/main/data/student-mat.csv"
↪  ,
header = TRUE,
sep = ",",
stringsAsFactors = FALSE
```

```r
)

student_por <- read.csv(
↵
↪  "https://raw.githubusercontent.com/Colochoo/cs4412-project/main/data/student-por.csv"↵
↪  ,
header = TRUE,
sep = ",",
stringsAsFactors = FALSE
)

# Add course identifier
student_math$course <- "Math"
student_por$course <- "Portuguese"

# Merge both datasets
student_data <- rbind(student_math, student_por)

# Check structure
str(student_data)
```

```
## 'data.frame':    1044 obs. of  34 variables:
##  $ school    : chr  "GP" "GP" "GP" "GP" ...
##  $ sex       : chr  "F" "F" "F" "F" ...
##  $ age       : int  18 17 15 15 16 16 16 17 15 15 ...
##  $ address   : chr  "U" "U" "U" "U" ...
##  $ famsize   : chr  "GT3" "GT3" "LE3" "GT3" ...
##  $ Pstatus   : chr  "A" "T" "T" "T" ...
##  $ Medu      : int  4 1 1 4 3 4 2 4 3 3 ...
##  $ Fedu      : int  4 1 1 2 3 3 2 4 2 4 ...
##  $ Mjob      : chr  "at_home" "at_home" "at_home" "health" ...
##  $ Fjob      : chr  "teacher" "other" "other" "services" ...
##  $ reason    : chr  "course" "course" "other" "home" ...
##  $ guardian  : chr  "mother" "father" "mother" "mother" ...
##  $ traveltime: int  2 1 1 1 1 1 1 2 1 1 ...
##  $ studytime : int  2 2 2 3 2 2 2 2 2 2 ...
##  $ failures  : int  0 0 3 0 0 0 0 0 0 0 ...
##  $ schoolsup : chr  "yes" "no" "yes" "no" ...
##  $ famsup    : chr  "no" "yes" "no" "yes" ...
##  $ paid      : chr  "no" "no" "yes" "yes" ...
##  $ activities: chr  "no" "no" "no" "yes" ...
##  $ nursery   : chr  "yes" "no" "yes" "yes" ...
##  $ higher    : chr  "yes" "yes" "yes" "yes" ...
##  $ internet  : chr  "no" "yes" "yes" "yes" ...
##  $ romantic  : chr  "no" "no" "no" "yes" ...
##  $ famrel    : int  4 5 4 3 4 5 4 4 4 5 ...
##  $ freetime  : int  3 3 3 2 3 4 4 1 2 5 ...
```

```
##  $ goout     : int  4 3 2 2 2 2 4 4 2 1 ...
##  $ Dalc      : int  1 1 2 1 1 1 1 1 1 1 ...
##  $ Walc      : int  1 1 3 1 2 2 1 1 1 1 ...
##  $ health    : int  3 3 3 5 5 5 3 1 1 5 ...
##  $ absences  : int  6 4 10 2 4 10 0 6 0 0 ...
##  $ G1        : int  5 5 7 15 6 15 12 6 16 14 ...
##  $ G2        : int  6 5 8 14 10 15 12 5 18 15 ...
##  $ G3        : int  6 6 10 15 10 15 11 6 19 15 ...
##  $ course    : chr  "Math" "Math" "Math" "Math" ...
```

```r
# Check Dimensions
dim(student_data)
```

```
## [1] 1044   34
```

One dataset was generated from the merge of the two datasets (students from Math Course plus the students from Portuguese Language Course), which outcome result in a total of 1044 observations (students) and 34 attributes. Also in this dataset we can see a variety of type of answers such as character nominal inputs in integer numbers format, character nominal inputs in binary format, and direct integer inputs.

Data Overview

```r
# Convert attributes to numeric
##Convert all yes and no answers to binary
student_data <- student_data %>%
  mutate(across(
    where(~ all(.x %in% c("yes", "no"))),
    ~ ifelse(.x == "yes", 1, 0)
  ))

##Convert nominal answers in binary format to binary
student_data <- student_data %>%
  mutate(
    sex = ifelse(sex == "M", 1, 0),
    address = ifelse(address == "R", 1, 0),
    famsize = ifelse(famsize == "GT3", 1, 0),
    Pstatus = ifelse(Pstatus == "T", 1, 0)
  )

##Convert nominal answers to numeric
student_data <- student_data %>%
  mutate(
    school = as.numeric(factor(school)),
```

```r
    Mjob = as.numeric(factor(Mjob)),
    Fjob = as.numeric(factor(Fjob)),
    reason = as.numeric(factor(reason)),
    guardian = as.numeric(factor(guardian)),
    course = as.numeric(factor(course))
  )

# Check structure in numeric form
str(student_data)
```

```
## 'data.frame':    1044 obs. of  34 variables:
##  $ school    : num  1 1 1 1 1 1 1 1 1 1 ...
##  $ sex       : num  0 0 0 0 0 1 1 0 1 1 ...
##  $ age       : int  18 17 15 15 16 16 16 17 15 15 ...
##  $ address   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ famsize   : num  1 1 0 1 1 0 0 1 0 1 ...
##  $ Pstatus   : num  0 1 1 1 1 1 1 0 0 1 ...
##  $ Medu      : int  4 1 1 4 3 4 2 4 3 3 ...
##  $ Fedu      : int  4 1 1 2 3 3 2 4 2 4 ...
##  $ Mjob      : num  1 1 1 2 3 4 3 3 4 3 ...
##  $ Fjob      : num  5 3 3 4 3 3 3 5 3 3 ...
##  $ reason    : num  1 1 3 2 2 4 2 2 2 2 ...
##  $ guardian  : num  2 1 2 2 1 2 2 2 2 2 ...
##  $ traveltime: int  2 1 1 1 1 1 1 2 1 1 ...
##  $ studytime : int  2 2 2 3 2 2 2 2 2 2 ...
##  $ failures  : int  0 0 3 0 0 0 0 0 0 0 ...
##  $ schoolsup : num  1 0 1 0 0 0 0 1 0 0 ...
##  $ famsup    : num  0 1 0 1 1 1 0 1 1 1 ...
##  $ paid      : num  0 0 1 1 1 1 0 0 1 1 ...
##  $ activities: num  0 0 0 1 0 1 0 0 0 1 ...
##  $ nursery   : num  1 0 1 1 1 1 1 1 1 1 ...
##  $ higher    : num  1 1 1 1 1 1 1 1 1 1 ...
##  $ internet  : num  0 1 1 1 0 1 1 0 1 1 ...
##  $ romantic  : num  0 0 0 1 0 0 0 0 0 0 ...
##  $ famrel    : int  4 5 4 3 4 5 4 4 4 5 ...
##  $ freetime  : int  3 3 3 2 3 4 4 1 2 5 ...
##  $ goout     : int  4 3 2 2 2 2 4 4 2 1 ...
##  $ Dalc      : int  1 1 2 1 1 1 1 1 1 1 ...
##  $ Walc      : int  1 1 3 1 2 2 1 1 1 1 ...
##  $ health    : int  3 3 3 5 5 5 3 1 1 5 ...
##  $ absences  : int  6 4 10 2 4 10 0 6 0 0 ...
##  $ G1        : int  5 5 7 15 6 15 12 6 16 14 ...
##  $ G2        : int  6 5 8 14 10 15 12 5 18 15 ...
##  $ G3        : int  6 6 10 15 10 15 11 6 19 15 ...
##  $ course    : num  1 1 1 1 1 1 1 1 1 1 ...
```

```r
# Column names (Attributes)
names(student_data)
```

```
##  [1] "school"     "sex"        "age"        "address"    "famsize"
##  [6] "Pstatus"    "Medu"       "Fedu"       "Mjob"       "Fjob"
## [11] "reason"     "guardian"   "traveltime" "studytime"  "failures"
## [16] "schoolsup"  "famsup"     "paid"       "activities" "nursery"
## [21] "higher"     "internet"   "romantic"   "famrel"     "freetime"
## [26] "goout"      "Dalc"       "Walc"       "health"     "absences"
## [31] "G1"         "G2"         "G3"         "course"
```

The combined dataset contains both Mathematics and Portuguese students and includes demographic, family, and academic variables that can be used for exploratory analysis and clustering. Several nominal variables were converted into numeric form for general preprocessing and visualization. However, nominal variables were not used directly in K-means clustering unless they represented meaningful ordinal or quantitative information, since arbitrary numeric codes can create misleading distances between categories.

Exploratory Data Analysis (EDA) Visualizations

Summary Statistics

```
##      school           sex              age           address
##  Min.   :1.000   Min.   :0.0000   Min.   :15.00   Min.   :0.000
##  1st Qu.:1.000   1st Qu.:0.0000   1st Qu.:16.00   1st Qu.:0.000
##  Median :1.000   Median :0.0000   Median :17.00   Median :0.000
##  Mean   :1.261   Mean   :0.4339   Mean   :16.73   Mean   :0.273
##  3rd Qu.:2.000   3rd Qu.:1.0000   3rd Qu.:18.00   3rd Qu.:1.000
##  Max.   :2.000   Max.   :1.0000   Max.   :22.00   Max.   :1.000
##     famsize          Pstatus            Medu            Fedu
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.000   Min.   :0.000
##  1st Qu.:0.0000   1st Qu.:1.0000   1st Qu.:2.000   1st Qu.:1.000
##  Median :1.0000   Median :1.0000   Median :3.000   Median :2.000
##  Mean   :0.7069   Mean   :0.8841   Mean   :2.603   Mean   :2.388
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:3.000
##  Max.   :1.0000   Max.   :1.0000   Max.   :4.000   Max.   :4.000
##       Mjob            Fjob           reason         guardian
##  Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:2.000   1st Qu.:3.000   1st Qu.:1.000   1st Qu.:2.000
##  Median :3.000   Median :3.000   Median :2.000   Median :2.000
##  Mean   :3.028   Mean   :3.246   Mean   :2.167   Mean   :1.837
##  3rd Qu.:4.000   3rd Qu.:4.000   3rd Qu.:3.000   3rd Qu.:2.000
##  Max.   :5.000   Max.   :5.000   Max.   :4.000   Max.   :3.000
##    traveltime      studytime       failures        schoolsup
##  Min.   :1.000   Min.   :1.00   Min.   :0.0000   Min.   :0.000
```

```
##  1st Qu.:1.000   1st Qu.:1.00   1st Qu.:0.0000   1st Qu.:0.000
##  Median :1.000   Median :2.00   Median :0.0000   Median :0.000
##  Mean   :1.523   Mean   :1.97   Mean   :0.2644   Mean   :0.114
##  3rd Qu.:2.000   3rd Qu.:2.00   3rd Qu.:0.0000   3rd Qu.:0.000
##  Max.   :4.000   Max.   :4.00   Max.   :3.0000   Max.   :1.000
##      famsup           paid          activities        nursery
##  Min.   :0.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:1.0000
##  Median :1.000   Median :0.0000   Median :0.0000   Median :1.0000
##  Mean   :0.613   Mean   :0.2107   Mean   :0.4943   Mean   :0.7998
##  3rd Qu.:1.000   3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:1.0000
##  Max.   :1.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
##      higher          internet         romantic          famrel
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :1.000
##  1st Qu.:1.0000   1st Qu.:1.0000   1st Qu.:0.0000   1st Qu.:4.000
##  Median :1.0000   Median :1.0000   Median :0.0000   Median :4.000
##  Mean   :0.9148   Mean   :0.7921   Mean   :0.3554   Mean   :3.936
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:5.000
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :5.000
##     freetime         goout            Dalc            Walc
##  Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:3.000   1st Qu.:2.000   1st Qu.:1.000   1st Qu.:1.000
##  Median :3.000   Median :3.000   Median :1.000   Median :2.000
##  Mean   :3.201   Mean   :3.156   Mean   :1.494   Mean   :2.284
##  3rd Qu.:4.000   3rd Qu.:4.000   3rd Qu.:2.000   3rd Qu.:3.000
##  Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000
##      health         absences            G1              G2
##  Min.   :1.000   Min.   : 0.000   Min.   : 0.00   Min.   : 0.00
##  1st Qu.:3.000   1st Qu.: 0.000   1st Qu.: 9.00   1st Qu.: 9.00
##  Median :4.000   Median : 2.000   Median :11.00   Median :11.00
##  Mean   :3.543   Mean   : 4.435   Mean   :11.21   Mean   :11.25
##  3rd Qu.:5.000   3rd Qu.: 6.000   3rd Qu.:13.00   3rd Qu.:13.00
##  Max.   :5.000   Max.   :75.000   Max.   :19.00   Max.   :19.00
##        G3            course
##  Min.   : 0.00   Min.   :1.000
##  1st Qu.:10.00   1st Qu.:1.000
##  Median :11.00   Median :2.000
##  Mean   :11.34   Mean   :1.622
##  3rd Qu.:14.00   3rd Qu.:2.000
##  Max.   :20.00   Max.   :2.000
```

Missing Values Check

```
##     school        sex        age    address    famsize    Pstatus       Medu
##          0          0          0          0          0          0          0
##       Fedu       Mjob       Fjob     reason   guardian traveltime  studytime
##          0          0          0          0          0          0          0
```

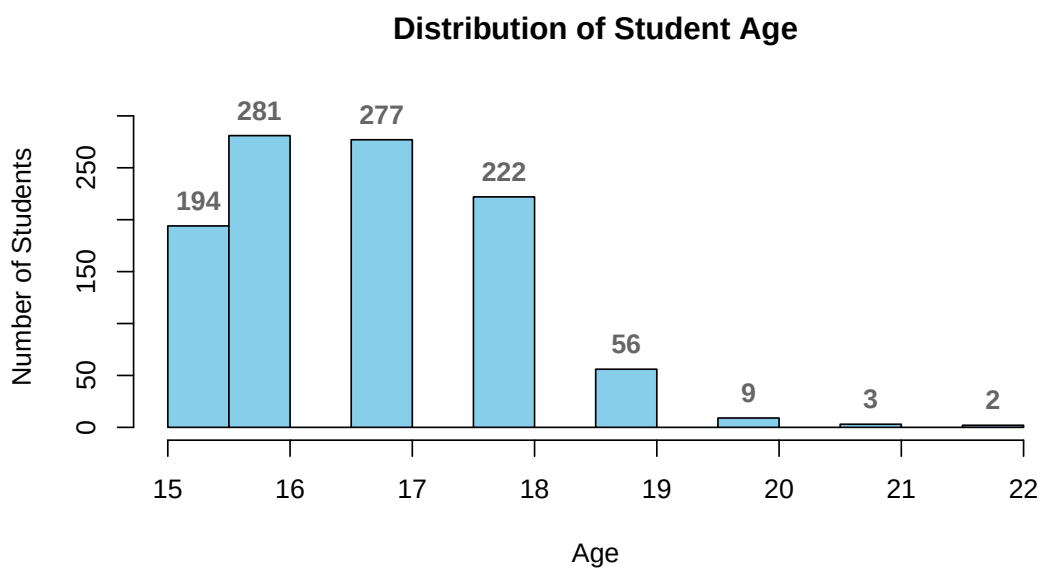
```
##   failures  schoolsup     famsup       paid activities    nursery     higher
##          0          0          0          0          0          0          0
##   internet   romantic     famrel   freetime      goout       Dalc       Walc
##          0          0          0          0          0          0          0
##     health   absences         G1         G2         G3     course
##          0          0          0          0          0          0
```

Duplicates Check

```
## [1] 0
```

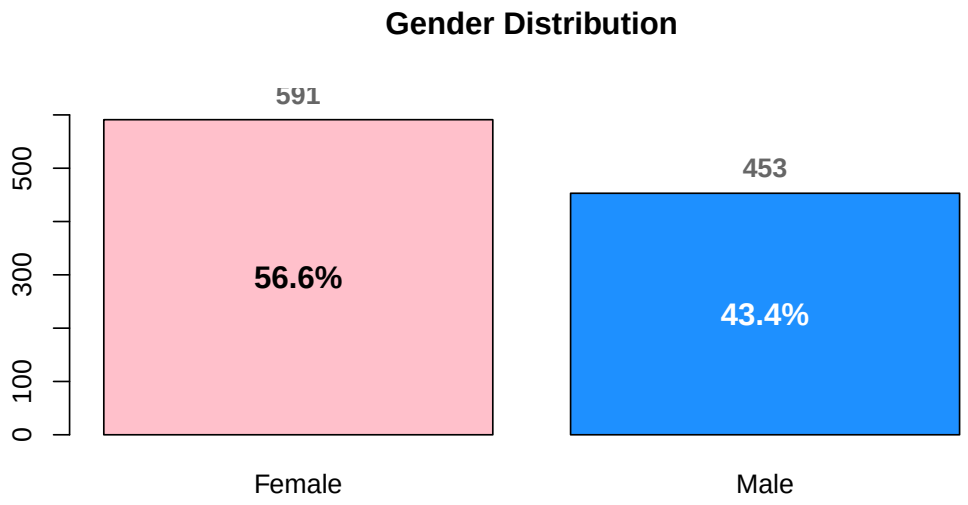
The dataset does not contain missing values or any duplicate value which simplifies preprocessing and makes the dataset suitable for clustering.

Age Distribution



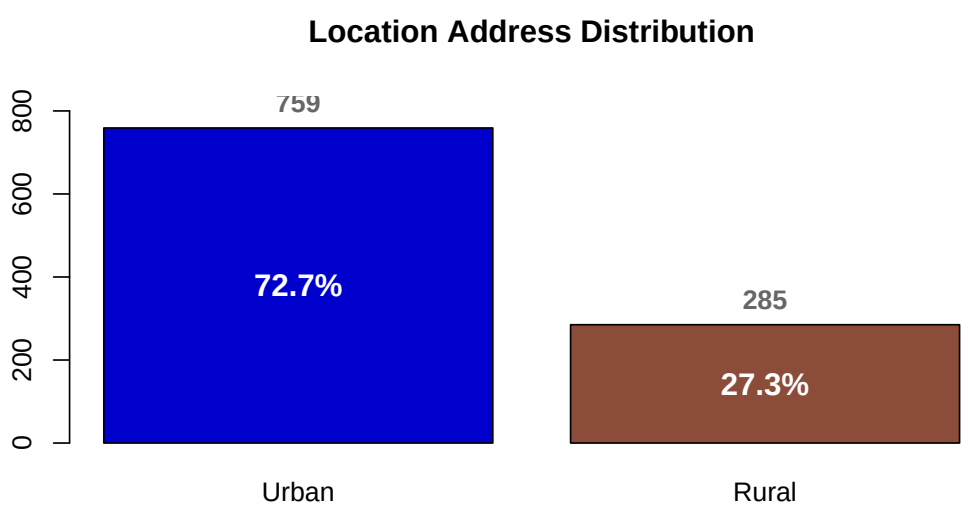
The histogram shows that most students are between ages 15 and 18. Very few students are older than 20, indicating the dataset primarily represents typical secondary school ages.

Gender Distribution



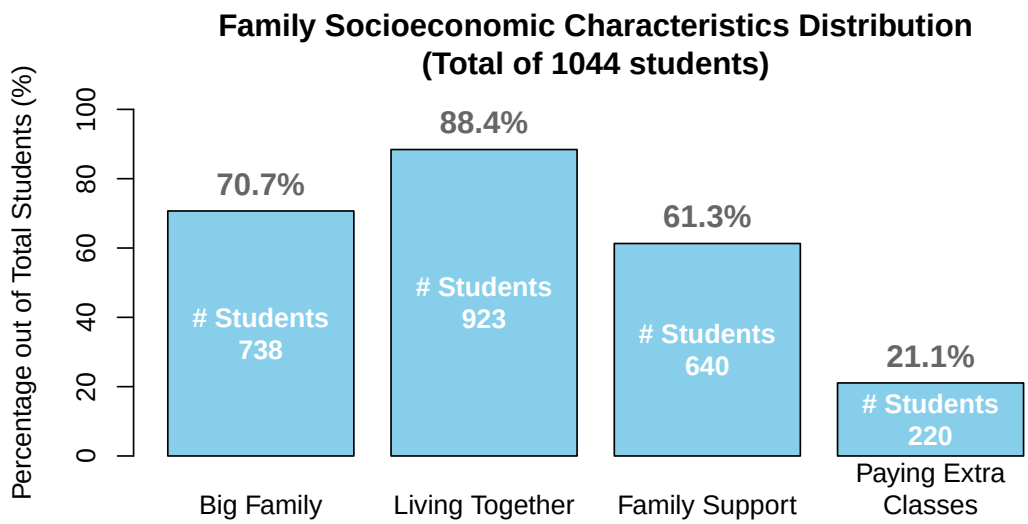
The dataset contains both male and female students, with females slightly more represented than males.

Student Home Address Distribution



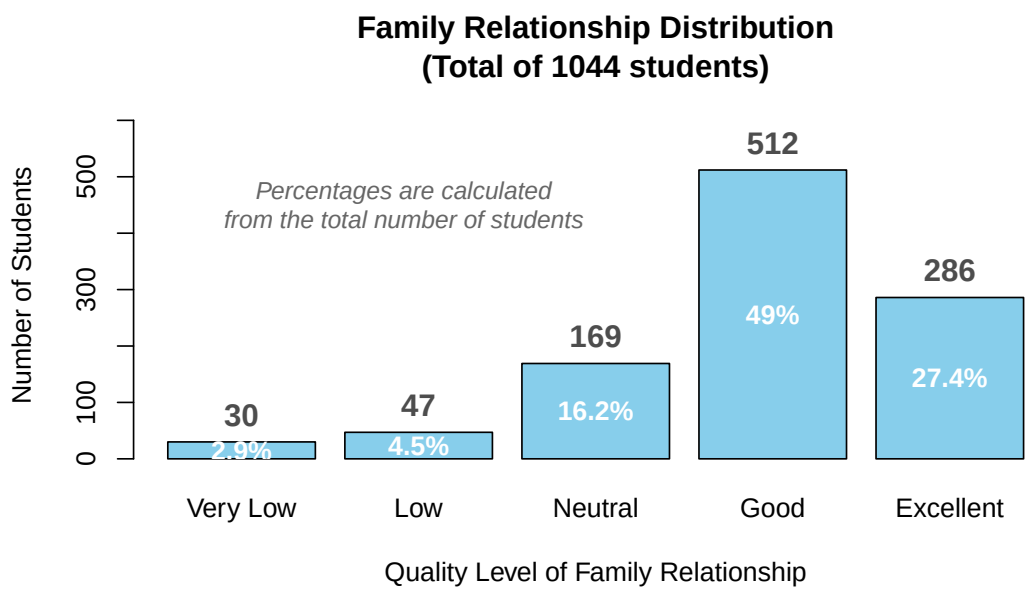
Most students in the dataset live in urban areas, while a smaller portion reside in rural areas. This indicates that the sample is predominantly composed of urban students.

Family Socioeconomic Distribution



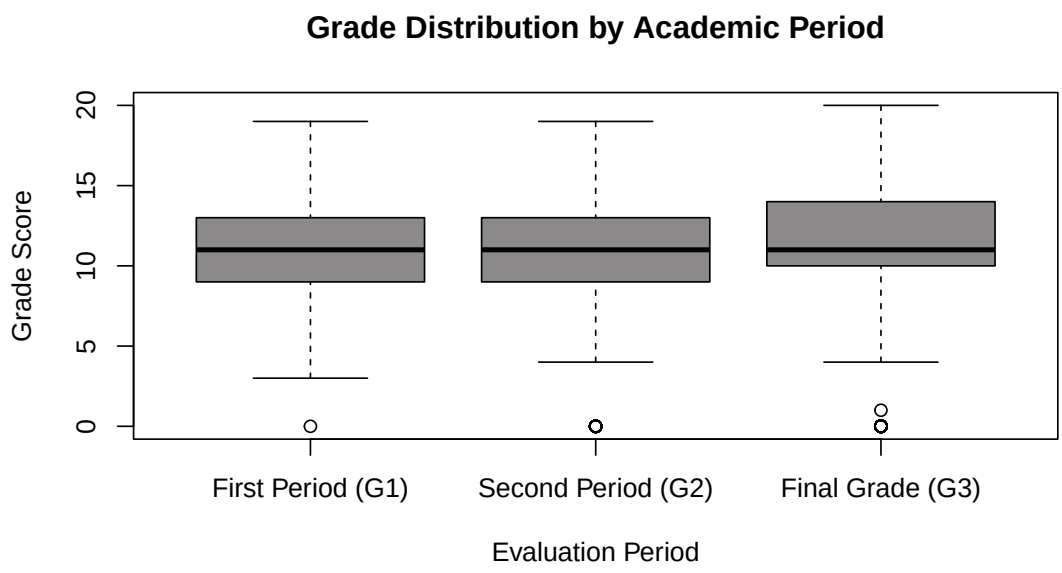
The family socioeconomic characteristics indicate that most students live with both parents. A majority also belong to larger families and report receiving family support. In contrast, only a small proportion of students attend paid extra classes. These findings suggest that while family support is relatively common, access to additional paid educational resources is limited for many students.

Family Relationship Distribution



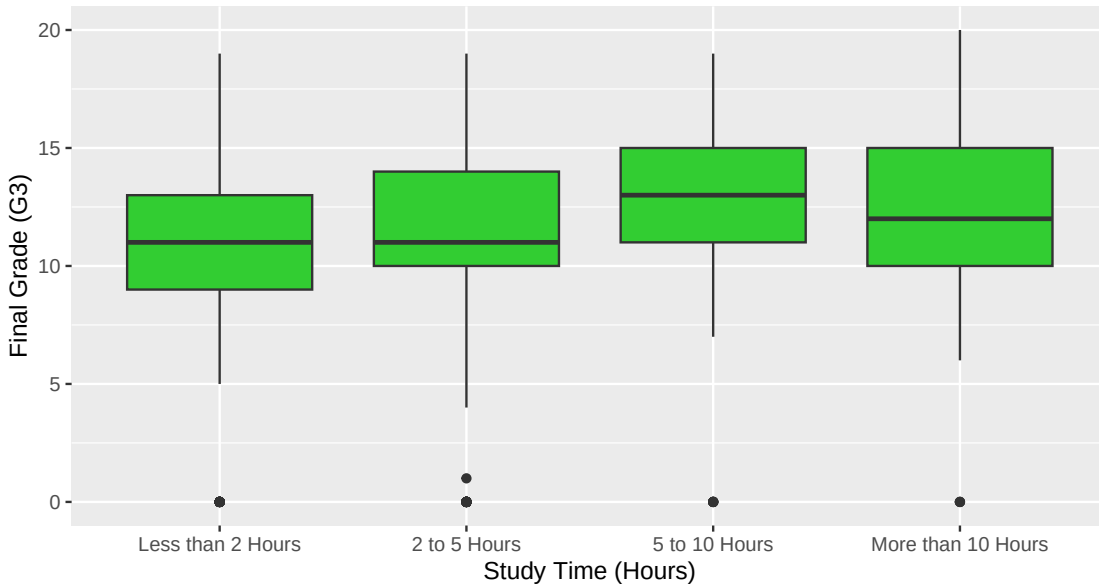
The distribution of family relationship quality shows that most students report positive family environments. More than half of the students rated their family relationship as 4 or 5, indicating strong family relationships for the majority of students. In contrast, very few students reported low relationship quality, with only less than 7.4% selecting levels 1 and 2. Overall, the data suggest that many students perceive their family relationships as supportive.

Grade Distribution Across Evaluation Periods



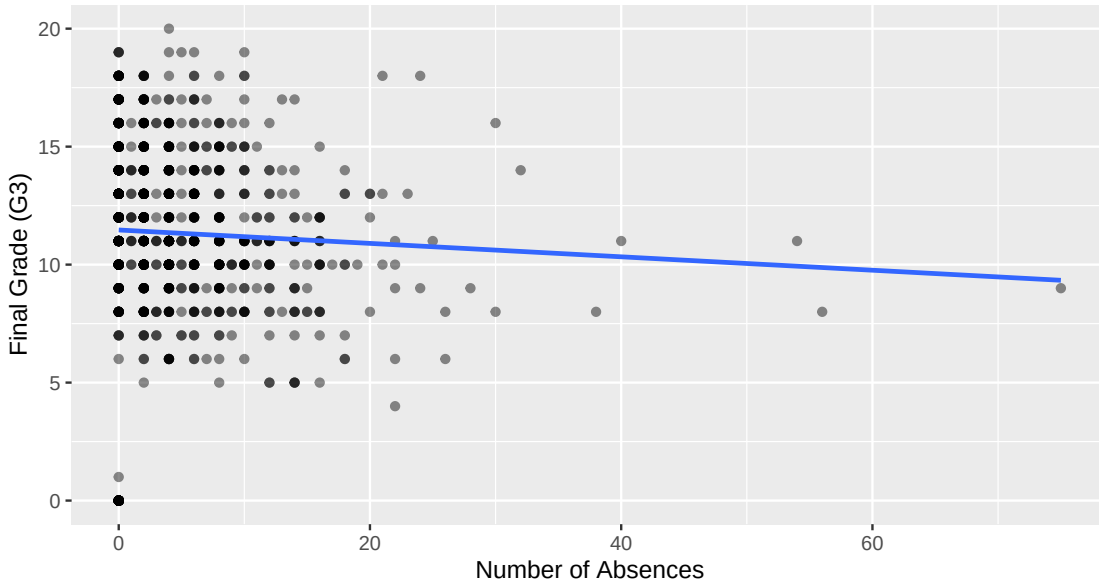
The boxplot shows that the median grade remains approximately constant across the three evaluation periods, indicating relatively stable academic performance over time. The spread of grades is also similar for G1, G2, and G3, with slightly greater variation in the final grade. Because G1 and G2 are sequential evaluation periods, the similarity to G3 suggests that early academic performance is a strong indicator of final outcomes.

Study Time vs Final Grade



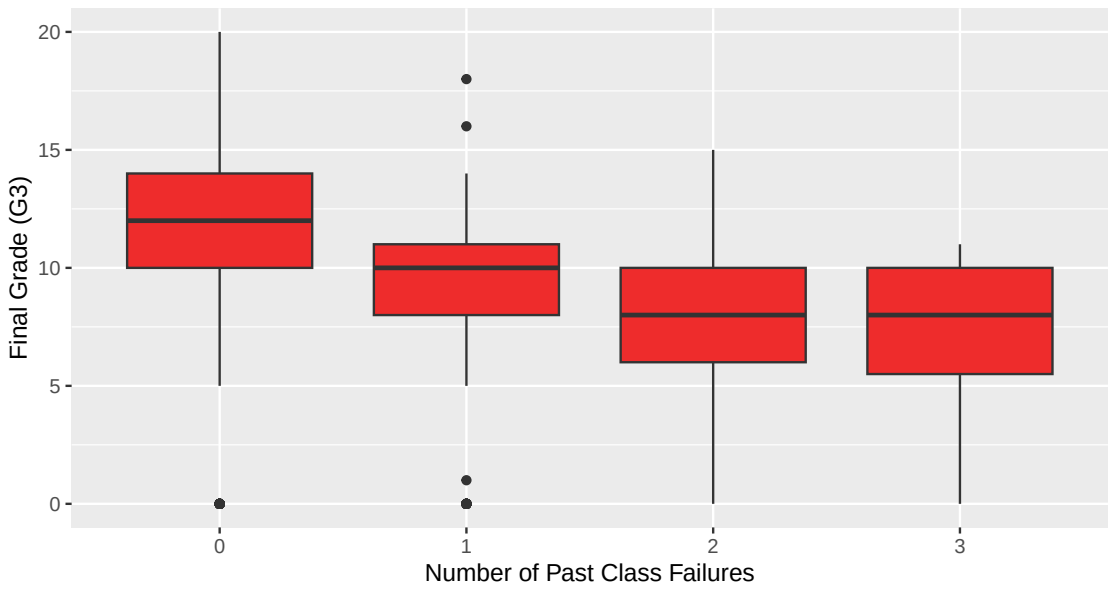
Students who spend more time studying tend to achieve higher final grades, though the relationship is not perfectly linear. The distribution indicates a modest upward trend between study time and final grade. Students in higher study-time categories tend to achieve somewhat better final grades, although there is still considerable overlap across groups. This suggests that study time contributes to performance, but it is not the only important factor.

Absences vs Final Grade



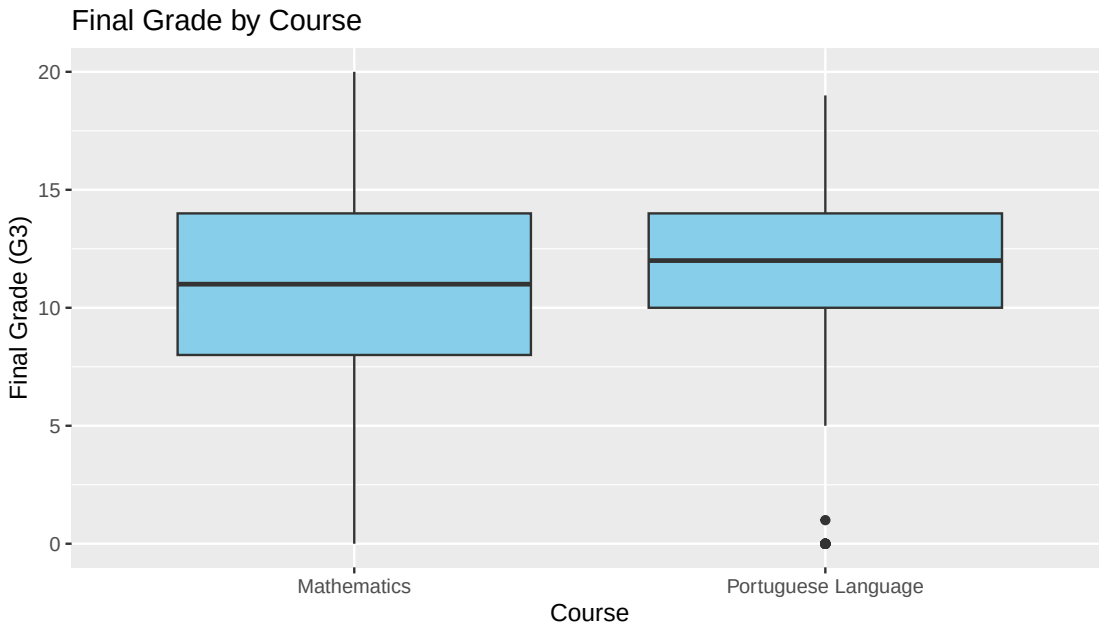
The scatterplot suggests a weak negative relationship between absences and final grade. Although some students with many absences still perform well, the overall trend indicates that more absences are generally associated with slightly lower academic performance.

Past Class Failures vs Final Grade



The number of past failures shows a clearer negative pattern than absences. Students with no prior failures tend to perform better, while students with more previous failures generally show lower final grades. This suggests that prior academic difficulty remains an important warning sign for later performance.

Course Comparison



Comparing Mathematics and Portuguese helps confirm whether the merged dataset behaves similarly across subjects. If the two distributions differ, course membership may matter when interpreting future clusters.

Correlation Analysis

Correlation Raw Data

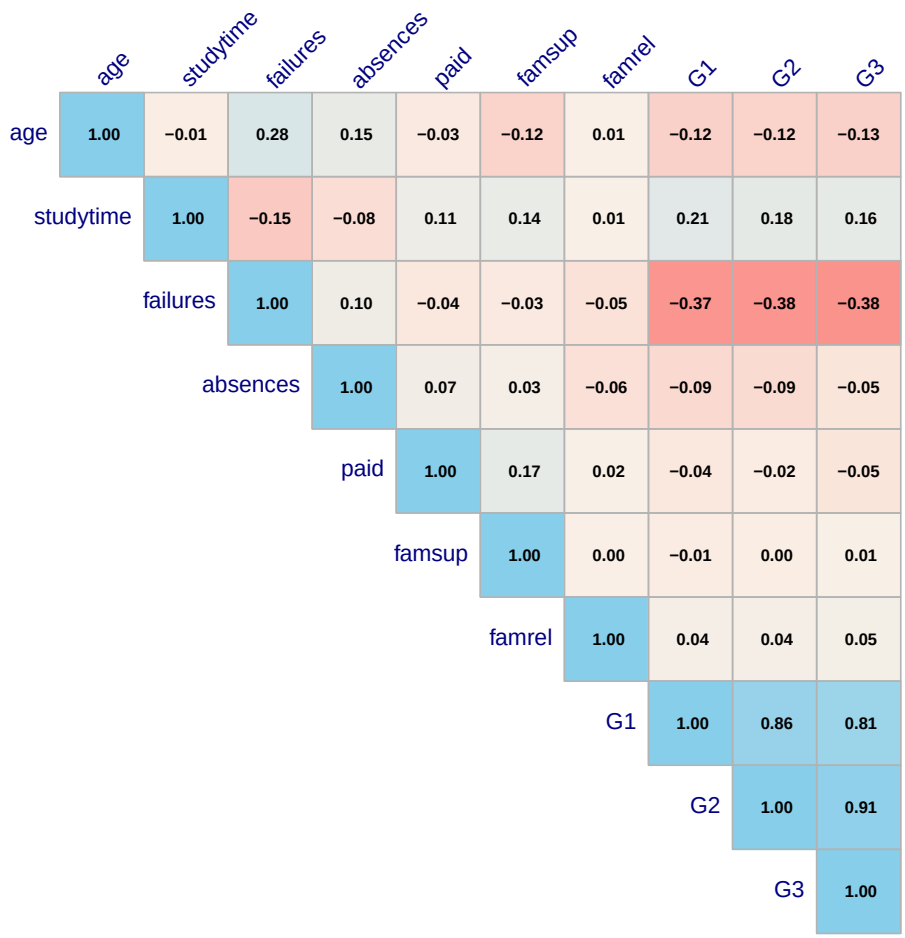
```
##                     age    studytime    failures    absences        paid
## age        1.000000000 -0.007870098  0.28236357  0.15319565 -0.02791748
## studytime -0.007870098  1.000000000 -0.15202352 -0.07559367  0.10570402
## failures   0.282363566 -0.152023523  1.00000000  0.09999785 -0.03638886
## absences   0.153195647 -0.075593669  0.09999785  1.00000000  0.07050534
## paid      -0.027917477  0.105704015 -0.03638886  0.07050534  1.00000000
## famsup    -0.116904384  0.143858194 -0.02757404  0.03126723  0.16942352
## famrel     0.007161921  0.012324093 -0.05367646 -0.06217066  0.01540385
## G1        -0.124121249  0.211313915 -0.37417487 -0.09242463 -0.03858786
## G2        -0.119474744  0.183166702 -0.37717218 -0.08933169 -0.01513367
## G3        -0.125282433  0.161628935 -0.38314528 -0.04567058 -0.04938811
##                 famsup       famrel          G1           G2          G3
## age       -0.116904384  0.007161921 -0.12412125 -0.119474744 -0.12528243
## studytime  0.143858194  0.012324093  0.21131391  0.183166702  0.16162894
## failures  -0.027574044 -0.053676457 -0.37417487 -0.377172176 -0.38314528
## absences   0.031267232 -0.062170662 -0.09242463 -0.089331694 -0.04567058
## paid       0.169423524  0.015403854 -0.03858786 -0.015133671 -0.04938811
## famsup     1.000000000  0.002261397 -0.01365500 -0.004520728  0.01331260
## famrel     0.002261397  1.000000000  0.03694727  0.042053621  0.05446106
```

```
## G1        -0.013654998  0.036947274  1.00000000  0.858738753  0.80914172
## G2        -0.004520728  0.042053621  0.85873875  1.000000000  0.91074316
## G3         0.013312604  0.054461059  0.80914172  0.910743163  1.00000000
```

Based on the correlation matrix, past academic results are strongly related to later grades. In particular, G1 is strongly correlated with G2 (0.85) and G3 (0.80), while G2 is very strongly correlated with G3 (0.91). This suggests that students who perform well in earlier periods tend to continue performing well in later evaluations.

Failure history has a moderate negative relationship with grades (approximately -0.37), indicating that students with more prior failures tend to earn lower grades. Study time shows only a weak positive relationship with grades, which means students who study more tend to perform slightly better on average, but the effect is not strong. Absences show only a weak negative relationship with performance in this dataset.

Correlation Visual Matrix Data

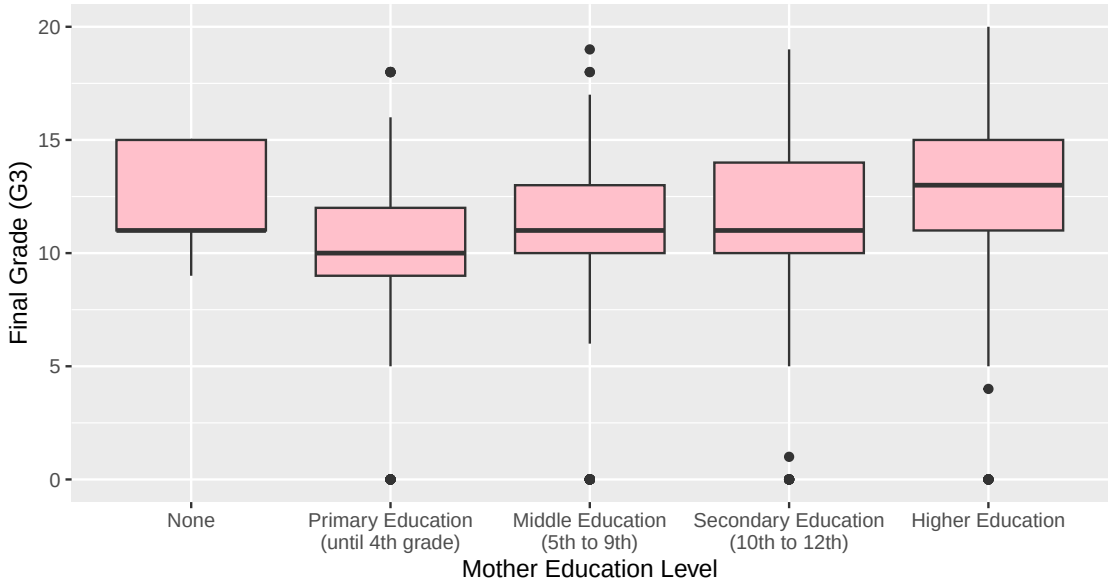


The correlation analysis shows that prior academic performance is the strongest predictor of final grade. In particular, G1 and G2 are strongly positively correlated with G3, indicating that students

who perform well early tend to maintain that performance later. Failures show a moderate negative relationship with grades, suggesting that students with a history of previous failures are more likely to earn lower final scores. Study time has only a weak positive relationship with grades, while absences show a weak negative relationship.

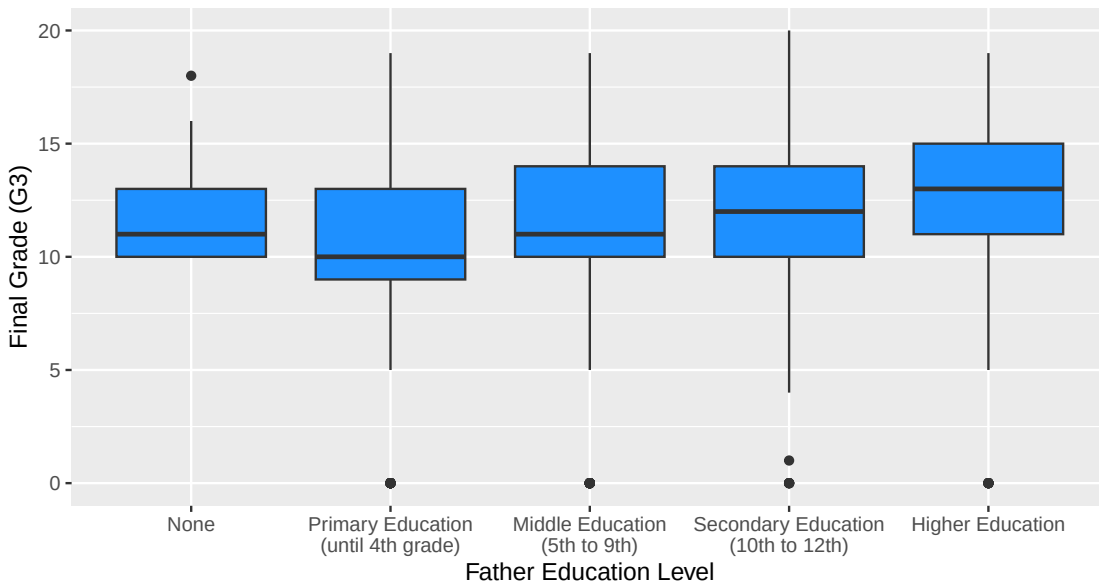
Socioeconomic Factors and Academic Performance

Mother's Education vs Final Grade



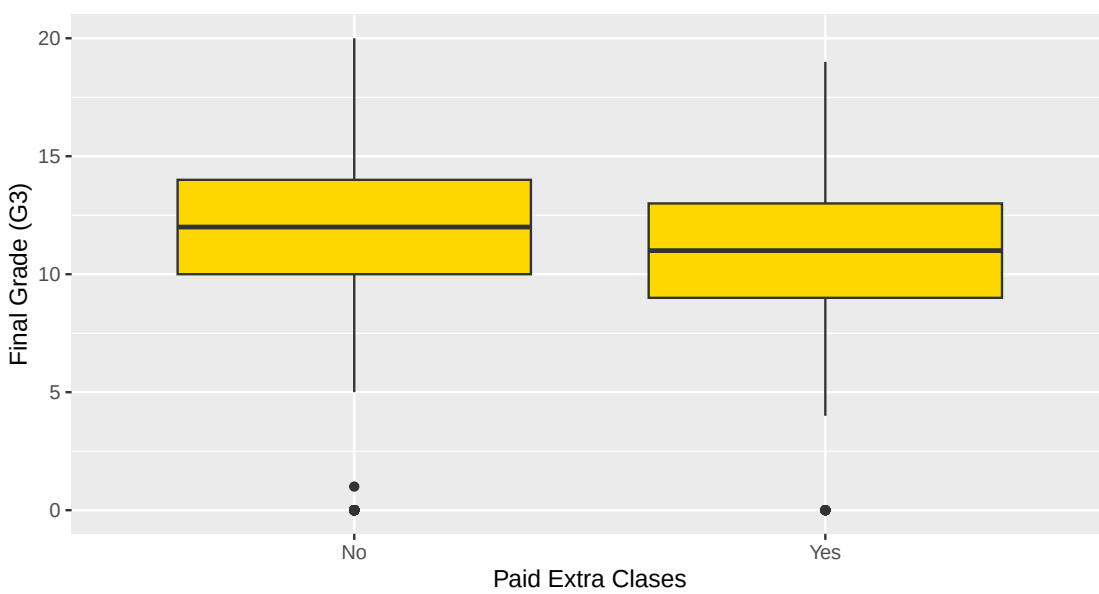
Students whose mothers have higher education levels tend to achieve slightly higher academic performance. This suggests that parental education may influence the level of academic support and learning environment available at home.

Father's Education vs Final Grade



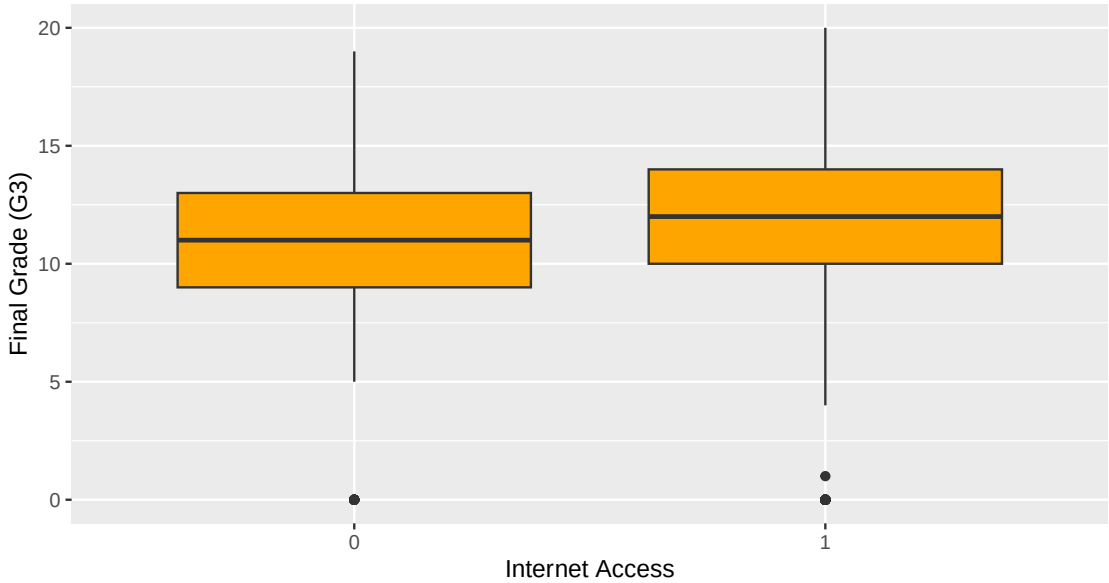
A similar pattern is observed with father's education level. Students with more educated parents tend to demonstrate somewhat greater academic consistency.

Paid Extra Classes vs Final Grade



The boxplot suggests that students who attend paid extra classes do not show a dramatically higher final grade distribution than those who do not. While there may be a slight difference in median performance, the overlap between the two groups is substantial. This indicates that paying for extra classes alone does not strongly distinguish academic outcomes in this dataset.

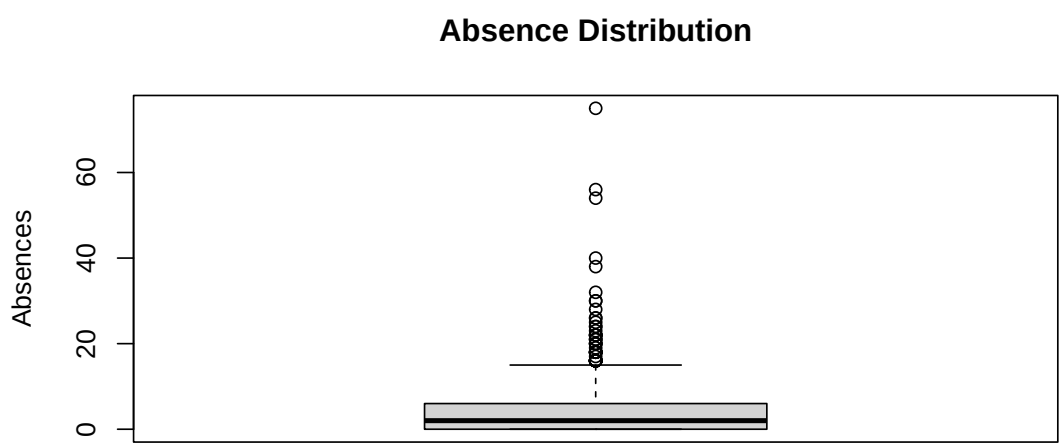
Internet Access vs Academic Performance

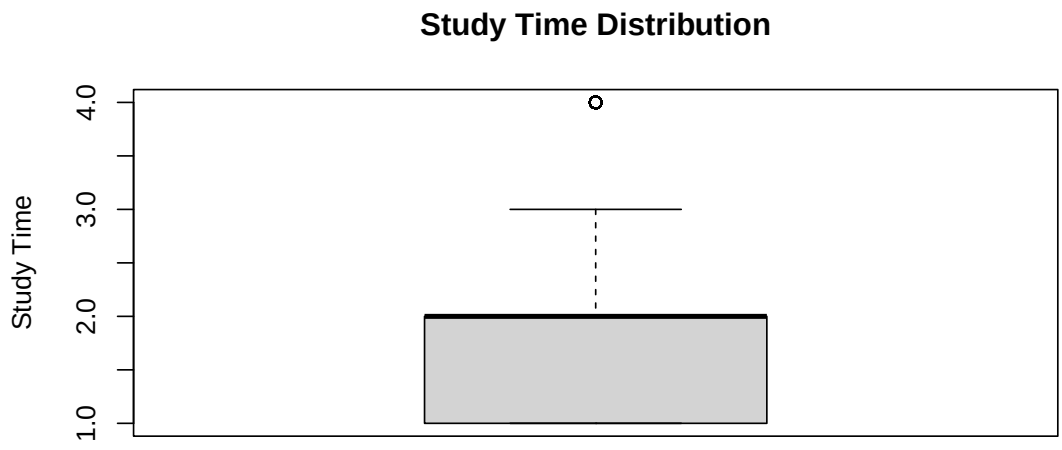


Students with internet access appear to have slightly higher median final grades than those without internet access. However, the distribution still overlap considerably, suggesting that internet access may provide an academic advantage but is not, by itself, a dominant predictor of performance.

Data Quality and Preprocessing Checks

Check selected variables for outliers oe spread





The dataset contains no duplicate rows. Some variables such as absences show a wider spread and possible outliers, while ordinal variables like studytime and freetime naturally take a limited set of values.

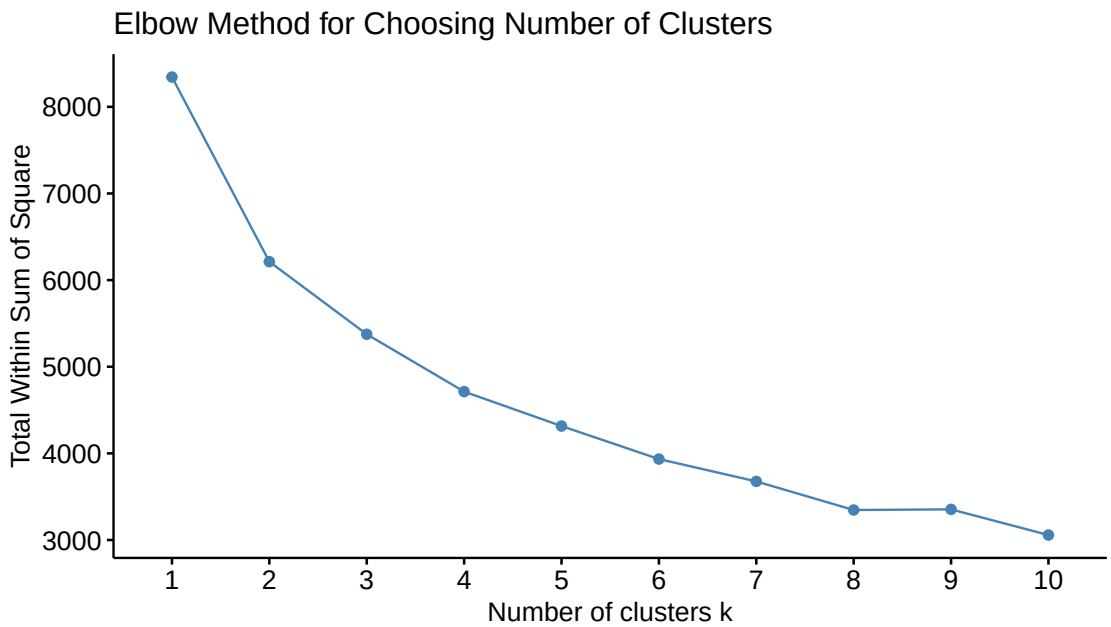
Clustering Preparation

The clustering analysis focuses on academic behavior and performance-related attributes. These variables were selected because they capture study habits, historical academic performance, and student engagement patterns. Although several nominal variables were numerically encoded earlier for preprocessing, the K-means model uses only scaled ordinal and quantitative variables: studytime,

failures, absences, G1, G2, G3, Medu, and Fedu. This helps avoid misleading numeric distances created by arbitrary coding of nominal categories.

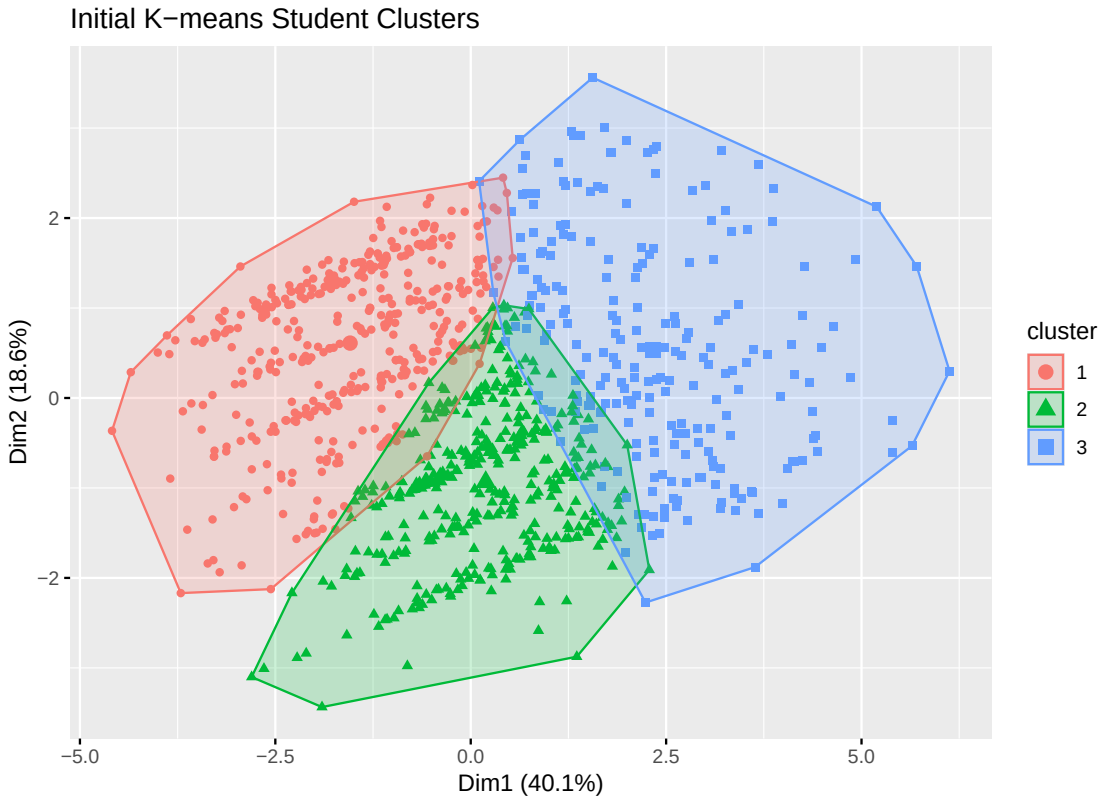
```
##    studytime          failures          absences             G1
##  Min.   :-1.16294   Min.   :-0.4029   Min.   :-0.7141   Min.   :-3.7587
##  1st Qu.:-1.16294   1st Qu.:-0.4029   1st Qu.:-0.7141   1st Qu.:-0.7420
##  Median : 0.03559   Median :-0.4029   Median :-0.3921   Median :-0.0716
##  Mean   : 0.00000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.: 0.03559   3rd Qu.:-0.4029   3rd Qu.: 0.2520   3rd Qu.: 0.5988
##  Max.   : 2.43266   Max.   : 4.1693   Max.   :11.3631   Max.   : 2.6099
##        G2                 G3                Medu              Fedu
##  Min.   :-3.42342   Min.   :-2.93468   Min.   :-2.3144   Min.   :-2.1710
##  1st Qu.:-0.68375   1st Qu.:-0.34723   1st Qu.:-0.5364   1st Qu.:-1.2618
##  Median :-0.07494   Median :-0.08848   Median : 0.3525   Median :-0.3527
##  Mean   : 0.00000   Mean   : 0.00000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.: 0.53388   3rd Qu.: 0.68776   3rd Qu.: 1.2415   3rd Qu.: 0.5565
##  Max.   : 2.36032   Max.   : 2.24023   Max.   : 1.2415   Max.   : 1.4656
```

Elbow Method



The elbow method helps identify a reasonable number of clusters by showing how the within cluster sum of squares decreases as the number of clusters increases. The point where the curve begins to flatten suggests a practical value for "k". Based on the elbow plot, k = 3 was selected because the reduction in within-cluster variation begins to level off after three clusters.

Initial K-means Clustering

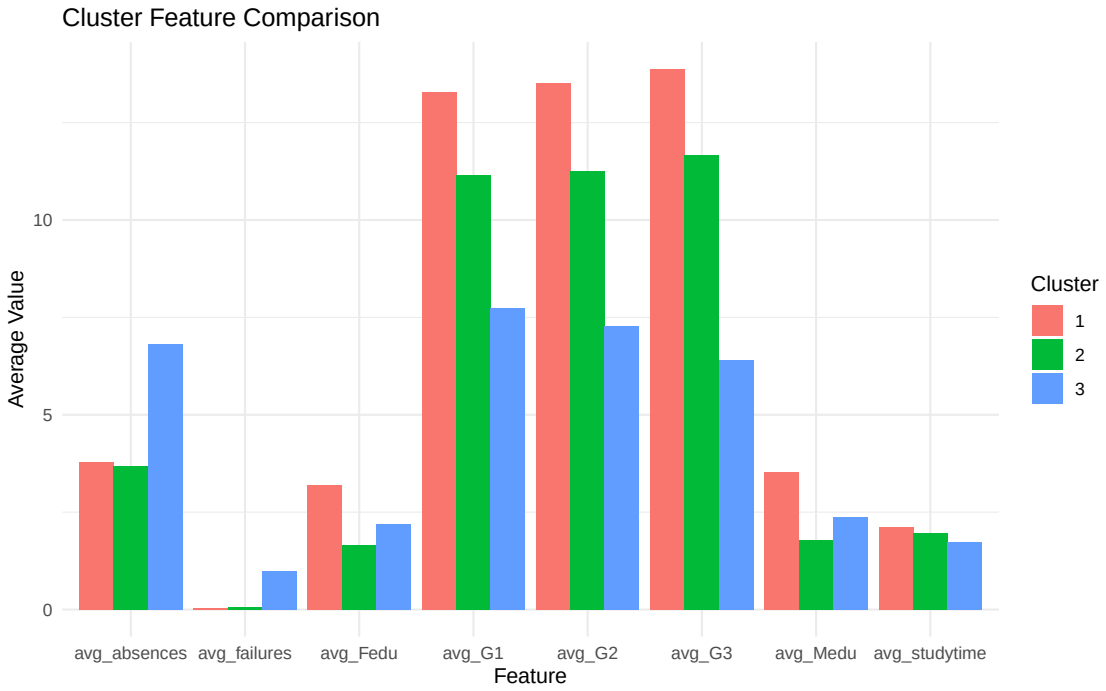


This initial clustering provides a first look at whether students form natural groups according to study behavior, prior performance, absences, and parental education. The three-cluster solution is intended to identify whether distinct student profiles emerge, such as higher-performing students, moderate-performing students, and students who may be more academically at risk.

Cluster Profile Summary

The cluster summary helps profile each group more clearly. Cluster 1 can be interpreted as a higher-performing group, with stronger earlier grades and very low failure history. Cluster 2 appears to represent moderate-performing students with average grades and relatively low failure levels. Cluster 3 appears to capture a more academically at-risk group, with the lowest average grades, the highest average failure history, and the highest average absences.

Cluster Feature Comparison

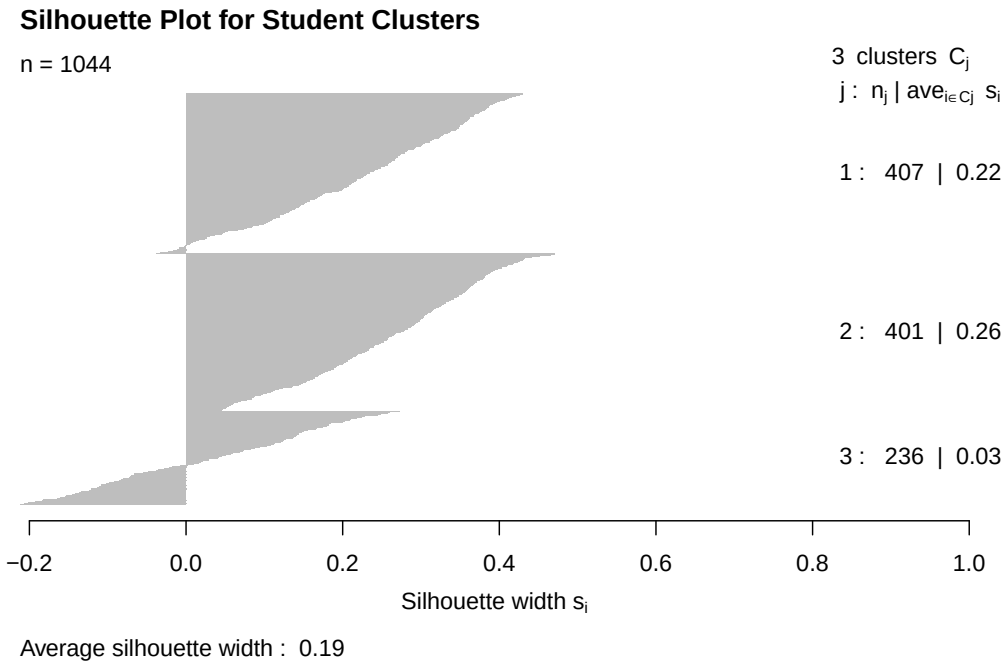


The cluster feature comparison visualization highlights the differences between the identified student groups across the selected variables. The chart shows how average study time, failures, absences, and grades vary between clusters. This reinforces the interpretation that one cluster represents higher-performing students with fewer failures, another reflects moderate academic performance, and a third cluster captures students at greater academic risk.

Silhouette Score Evaluation

```
## [1] 0.1934213
```

To evaluate the quality of the clustering solution, a silhouette analysis was performed. The silhouette score measures how similar each observation is to its own cluster compared to other clusters. Values range from -1 to 1, where higher values indicate better separation between clusters. The average silhouette score for this clustering solution provides an additional indication of how well the student groups are separated in the feature space.



Because G1, G2, and G3 are highly correlated variables, they strongly influence the clustering outcome. Future work could explore dimensionality reduction methods such as Principal Component Analysis (PCA) to reduce redundancy and better capture underlying student patterns.

Pattern Discovery Results

Discovery Question 1: Family and Socioeconomic Patterns

The analysis suggests that family and socioeconomic characteristics do show recognizable patterns, although their relationship with academic performance is generally modest compared with direct academic indicators. Family relationship quality is generally positive across the sample, and parental education appears to have a slight positive association with final grades. In contrast, paid extra classes do not strongly distinguish higher-performing students from lower-performing students, while internet access may provide a small advantage but is not a dominant predictor on its own.

Discovery Question 2: Study Habits, Absences, and Performance

Study habits and absences show some relationship with academic performance, but the patterns are not equally strong. Study time has a weak positive relationship with final grade, meaning that students who study more tend to perform slightly better on average. Absences show a weak negative relationship with performance, while previous failures show a much more noticeable negative pattern. This suggests that long-term academic history is a stronger signal of student risk than attendance alone.

Discovery Question 3: Natural Student Clusters

The K-means clustering results suggest that three natural student groups emerge when combining academic, behavioral, and family-education variables. One cluster represents higher-performing students with stronger grade histories and fewer failures, another captures moderate-performing students, and a third reflects a more academically at-risk group with lower grades, more failures, and higher absences. These results suggest that student performance patterns are driven more strongly by academic history than by isolated socioeconomic variables.

Key Findings

1. The dataset is complete and suitable for exploratory analysis and clustering.
2. Prior grades, especially G1 and G2, are the strongest predictors of final performance.
3. Past failures show a more noticeable negative relationship with final grade than absences.
4. Study time has a positive but relatively weak relationship with final grade.
5. Family relationships are generally positive across the sample, and parental education appears to have a modest association with performance.
6. The K-means solution suggests three student profiles: higher-performing, moderate-performing, and academically at-risk.

Conclusion

This report provides an initial exploratory and preprocessing foundation for the Student Performance project. The visualizations and correlation analysis suggest that previous academic outcomes and failure history are more strongly associated with final grades than isolated socioeconomic variables. The initial clustering setup also offers a starting point for identifying natural student profiles. In future milestones, the clustering results can be evaluated more formally and expanded with additional interpretation, validation metrics, and predictive modeling.